Lecture 7: Fixed Income Markets

Questions

These five multiple-choice questions follow the lecture's reasoning with different numbers. Work through the problem before reading the solutions. Keep full precision until the final answer.

All bonds pay semiannually. Assume no default, taxes, fees or early redemption; value on a coupon date. Annual YTM uses semiannual compounding.

1. Read the bond contract

A bond has $2,000 face value, an 8% annual coupon, two years to maturity and semiannual payments. Which is its complete future cash-flow sequence?

A. $80, $80, $80, $2,000

B. $160, $160, $160, $2,160

C. $80, $80, $80, $2,080

D. $40, $40, $40, $2,040

2. Price the same bond

Use the bond in Question 1 and a quoted annual YTM of 10%, compounded semiannually. What price should an investor pay?

A. $1,929.08

B. $2,000.00

C. $2,074.34

D. $1,859.16

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3. Recover the annual YTM

The bond in Question 1 sells for $1,929.080989916. Which Excel calculation reports the quoted annual YTM?

A. =2*RATE(2,160,-1929.080989916,2000); 10%

B. =80/1929.080989916; 4.15%

C. =RATE(4,80,-1929.080989916,2000); 5%

D. =2*RATE(4,80,-1929.080989916,2000); 10%

4. Compare premium and discount prices

Keep the same 8% coupon bond. What are its classifications at market YTMs of 6%, 8% and 10%, respectively?

A. Premium; discount; par

B. Par; premium; discount

C. Premium; par; discount

D. Discount; par; premium

5. Interpret a credit spread

A corporate bond yields 6.25%; a Treasury with comparable maturity and the same quoting convention yields 4.50%. What is the spread, and what does it mean?

A. 175 basis points; compensation may reflect credit and liquidity risk

B. 1.75 basis points; a guaranteed excess return

C. 175 basis points; the corporate bond cannot default

D. 625 basis points; compensation only for inflation

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Worked solutions

1. Read the bond contract - Answer C

Step 1. Find the coupon. I apply the annual coupon rate to face value, not to price: 2,000 x 8% = $160 per year. Two payments give $80 each.

Step 2. Count payment dates. Two years times two payments per year gives four periods.

Step 3. Add principal once. The last payment contains both the $80 coupon and $2,000 principal.

$$C = 2000(0.08)/2 = 80, \quad n = 2(2) = 4$$

Interpretation. The contractual cash flows are known before we choose the discount rate.

Check your reasoning. Do not count the principal twice or omit the final coupon.

2. Price the same bond - Answer A

Step 1. Match the period. The six-month discount rate is 10% / 2 = 5%. There are four half-year periods.

Step 2. Value coupons and principal. Coupon PV is 283.6760; principal PV is 1,645.4049.

Step 3. Add present values. The total is $1,929.080990, or $1,929.08 after rounding.

$$P = 80\frac{1 - 1.05^{-4}}{0.05} + 2000(1.05)^{-4} = 1929.080990$$

Interpretation. The 8% coupon is below the 10% market yield, so this bond trades at a discount.

Check your reasoning. Using 10% every six months doubles the periodic rate.

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3. Recover the annual YTM - Answer D

Step 1. Set the signs. The purchase price is a cash outflow; coupons and principal are receipts.

Step 2. Use the contract periods. RATE uses four periods, an $80 payment and $2,000 face value. It returns 5% per semester.

Step 3. Annualize the quote. Multiply that semester yield by two to report 10% nominal annual YTM.

$$y = 2\ \mathrm{RATE}(4, 80, -1929.080989916, 2000) = 10\%$$

Interpretation. Current yield uses annual coupons divided by price; it does not solve the full present-value equation.

Check your reasoning. Do not put $2,080 in the final-value argument: RATE already includes the final coupon.

4. Compare premium and discount prices - Answer C

Step 1. Keep cash flows fixed. Only the required yield changes; the coupon remains $80 each semester.

Step 2. Compare coupon and yield. At 6%, the coupon is above the required yield. At 8%, they match. At 10%, the coupon is below the yield.

Step 3. Check the prices. The respective prices are $2,074.34, $2,000.00 and $1,929.08.

$$c > y \Rightarrow P > FV; \quad c = y \Rightarrow P = FV; \quad c < y \Rightarrow P < FV$$

Interpretation. Higher required returns reduce the price of the same future payments.

Check your reasoning. The coupon does not reset when the YTM changes.

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5. Interpret a credit spread - Answer A

Step 1. Subtract comparable yields. 6.25% - 4.50% = 1.75 percentage points.

Step 2. Convert units. One percentage point is 100 basis points; 1.75 x 100 = 175 basis points.

Step 3. Interpret the promise. The quoted yield assumes contractual payments. Credit losses or selling early can change the realized return.

$$\text{Spread} = (0.0625 - 0.0450) \times 10000 = 175 \text{ bp}$$

Interpretation. A spread is a pricing difference, not a guarantee of additional realized return.

Check your reasoning. A rating does not eliminate default risk.